Кладковой Максим 5130904/30005 Вар 47

Интерполяционный кубический сплайн

$f(x) = \sin(\pi x)$; $x \in \left[-\frac{1}{4}; \frac{1}{4}\right]$ $n=3$

№	x	$f(x)$	$\frac{df}{dx} = m$
0	$-\frac{1}{4}$	$-\frac{\sqrt{2}}{2}$	$\frac{\pi\sqrt{2}}{2} = m_0$
1	$-\frac{1}{6}$	$-\frac{1}{2}$	$\frac{\pi\sqrt{3}}{2} = m_1$
2	0	0	$\pi = m_2$
3	$\frac{1}{4}$	$\frac{\sqrt{2}}{2}$	$\frac{\pi\sqrt{2}}{2} = m_3$

$$\begin{cases} \frac{m_0}{h_1} + 2\left(\frac{1}{h_1} + \frac{1}{h_2}\right)m_1 - \frac{m_2}{h_2} = \frac{3(f_2 - f_1)}{h_2^2} + \frac{3(f_1 - f_0)}{h_1^2} \\ \frac{m_1}{h_2} + 2\left(\frac{1}{h_2} + \frac{1}{h_3}\right)m_2 + \frac{m_3}{h_3} = \frac{3(f_3 - f_2)}{h_3^2} + \frac{3(f_2 - f_1)}{h_2^2} \end{cases}$$

$$\begin{cases} 2\left(\frac{1}{h_1} + \frac{1}{h_2}\right)m_1 + \frac{m_2}{h_2} = \frac{3(f_2 - f_1)}{h_2^2} + \frac{3(f_1 - f_0)}{h_1^2} - \frac{f_0'}{h_1} \\ \frac{m_1}{h_2} + 2\left(\frac{1}{h_2} + \frac{1}{h_3}\right)m_2 = \frac{3(f_3 - f_2)}{h_3^2} + \frac{3(f_2 - f_1)}{h_2^2} - \frac{f_3'}{h_3} \end{cases}$$

$$h_1 = \frac{1}{12}; h_2 = \frac{1}{6}; h_3 = \frac{1}{4}$$

$$\begin{cases} 36m_1 + 6m_2 = 108(f_2 - f_1) + 432(f_1 - f_0) - 6\pi\sqrt{2} \\ 6m_1 + 20m_2 = 48(f_3 - f_2) + 108(f_2 - f_1) - 2\pi\sqrt{2} \end{cases}$$

$$\begin{cases} 36m_1 + 6m_2 = 108\left(\frac{1}{2}\right) + 432\left(-\frac{1}{2} + \frac{\sqrt{2}}{2}\right) - 6\pi\sqrt{2} \\ 6m_1 + 20m_2 = 48\left(\frac{\sqrt{2}}{2}\right) + 108\left(\frac{1}{2}\right) - 2\pi\sqrt{2} \end{cases}$$

$$\begin{cases} 6m_1 = 24\sqrt{2} + 54 - 2\pi\sqrt{2} - 20m_2 \\ 6(24\sqrt{2} + 54 - 2\pi\sqrt{2} - 20m_2) = 54 - 216 + 216\sqrt{2} - 6\pi\sqrt{2} \end{cases}$$

$$m_1 = \frac{-99 + 116\sqrt{2} - 3\pi\sqrt{2}}{19} \approx 2,72211$$

$$m_2 = \frac{-12\sqrt{2} + 81 - \pi\sqrt{2}}{19} \approx 3,13613$$

$$H_i(t) = f_{i-1}(1 - 3t^2 + 2t^3) + f_i(3t^2 - 2t^3) + m_{i-1}h_i(t - 2t^2 + t^3) + m_i h_i(-t^2 + t^3)$$

$$\frac{dH_i(t)}{dx} = \frac{6(f_i - f_{i-1})}{h_i}(t - t^2) + m_{i-1}(1 - 4t + 3t^2) + m_i(-2t + 3t^2)$$

$$\frac{d^2H_i(t)}{dx^2} = \frac{6(f_i - f_{i-1})}{h_i^2}(1 - 2t) + \frac{m_{i-1}(-4 + 6t)}{h_i} + \frac{m_i(-2 + 6t)}{h_i}$$

$$\frac{d^3H_i(t)}{dx^3} = -\frac{12(f_i - f_{i-1})}{h_i^3} + \frac{6(m_i + m_{i-1})}{h_i^2}$$

$$1)\ h_1=\frac{1}{12};\ t=12x+3;\ x\in\left[-\frac{1}{4};-\frac{1}{6}\right]$$

$$H_1(t)=-\frac{\sqrt{2}}{2}\left(1-3(12x+3)^2+2(12x+3)^3\right)-\frac{1}{2}\left(3(12x+3)^2-2(12x+3)^3\right)+\frac{\pi\sqrt{2}}{24}\left(12x+3-2(12x+3)^2+\right.$$

$$\left.+(12x+3)^3\right)+2{,}72211\cdot\frac{1}{12}\left(-(12x+3)^2+(12x+3)^3\right)$$

$$\frac{dH_1(t)}{dx}=\frac{6\left(-\frac{1}{2}+\frac{\sqrt{2}}{2}\right)}{\frac{1}{12}}\left(12x+3-(12x+3)^2\right)+\frac{\pi\sqrt{2}}{2}\left(1-4(12x+3)+3(12x+3)^2\right)+2{,}72211\left(-2(12x+3)+3(12x\ldots\right.$$

$$\frac{d^2H_1(t)}{d^2x}=144\left(6\left(-\frac{1}{2}+\frac{\sqrt{2}}{2}\right)\left(1-2(12x+3)\right)+6\pi\sqrt{2}\left(-4+6(12x+3)\right)+12\cdot 2{,}72211\left(-2+6(12x+3)\right)\right)$$

$$\frac{d^3H_1(t)}{d^3x}=1728(-12)\left(-\frac{1}{2}+\frac{\sqrt{2}}{2}\right)+144\cdot 6\left(\frac{\pi\sqrt{2}}{2}+2{,}72211\right)$$

$$r_1(x)=f(x)-H_1(x)=\sin(\pi x)+\frac{\sqrt{2}}{2}\left(1-3(12x+3)^2+2(12x+3)^3\right)+\frac{1}{2}\left(3(12x+3)^2-2(12x+3)^3\right)-\frac{\pi\sqrt{2}}{24}\cdot$$

$$\cdot\left(12x+3-2(12x+3)^2+(12x+3)^3\right)-\frac{2{,}72211}{12}\left(-(12x+3)^2+(12x+3)^3\right)=\sin(\pi x)+x^3(1728\sqrt{2}$$

$$-1728-72\pi\sqrt{2}-391{,}98384)+x^2(1080\sqrt{2}-1080-42\pi\sqrt{2}-261{,}32256)+x(216\sqrt{2}-216-$$

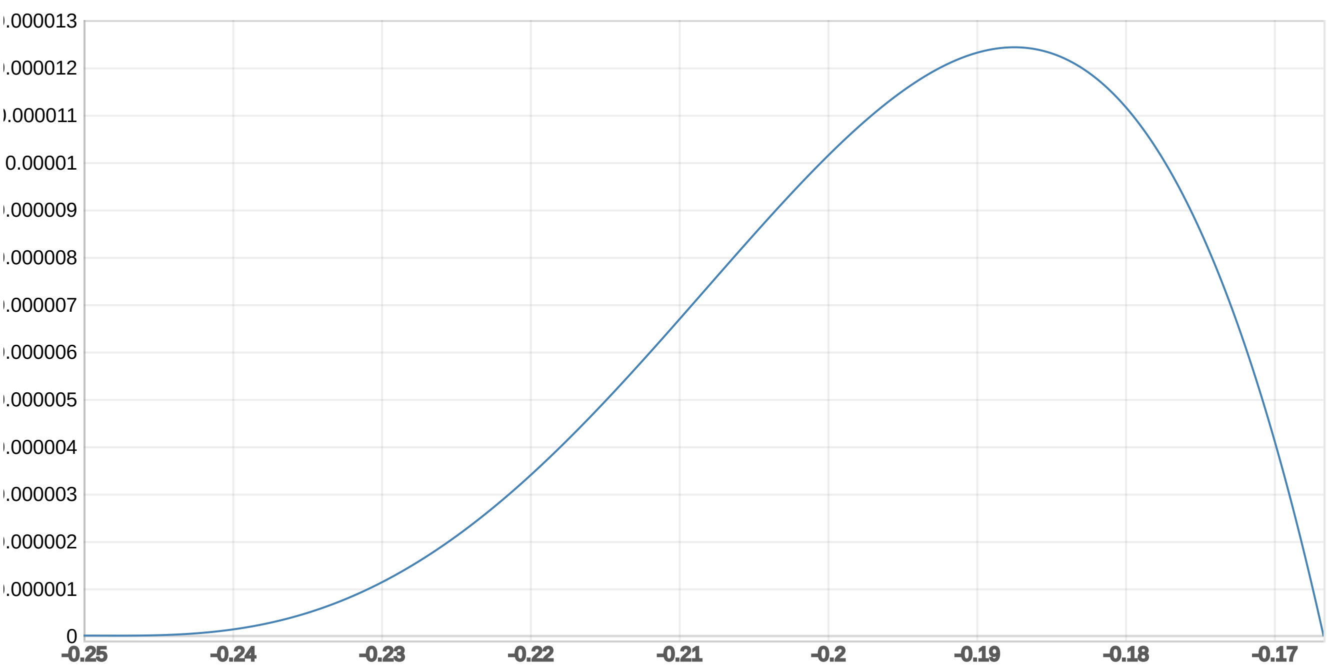
$$-8\pi\sqrt{2}-57{,}16431)+\left(14\sqrt{2}-\frac{27}{2}-\frac{\pi\sqrt{2}}{2}-4{,}08316\right)$$

$$\frac{dr_1(x)}{dx}=\pi\cos(\pi x)+3x^2(1728\sqrt{2}-2119{,}98384-72\pi\sqrt{2})+2x(1080\sqrt{2}-1341{,}32256-42\pi\sqrt{2})+$$

$$+(216\sqrt{2}-273{,}16431-8\pi\sqrt{2})$$

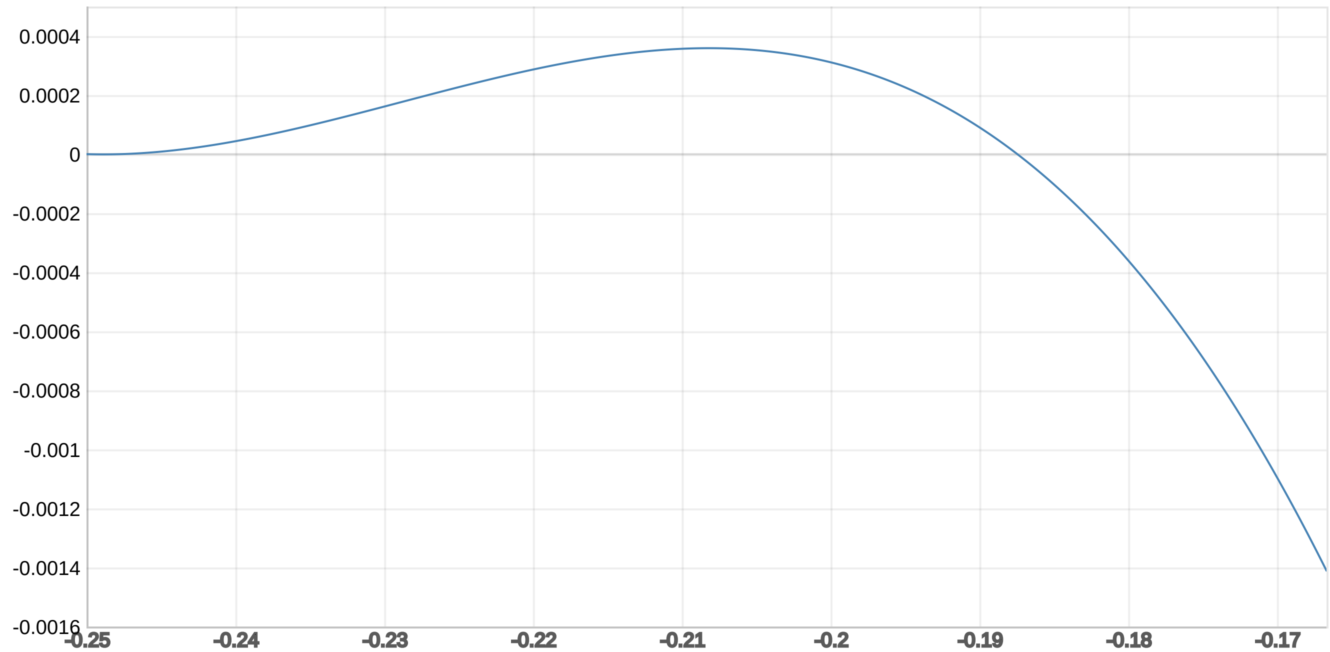
$r_1(x)$:

$$\left[3.8896235\cdot x^3-0.572996391\cdot x^2-3.2372441\cdot x+\sin\left(3.1415926536\times 10^0\cdot x\right)-5.616599\times 10^{-3}\right]$$



$\frac{dr_1(x)}{dx}$:

$$\left[1.16688705\times 10^1\cdot x^2-1.145992782\cdot x+3.1415926536\times 10^0\cdot\cos\left(3.1415926536\times 10^0\cdot x\right)-3.2372441\right]$$



2) $h_2 = \frac{1}{6}$; $t = 6x+1$; $x \in [-\frac{1}{6}; 0]$

$$H_2(t) = -\frac{1}{2}(1-3(6x+1)^2+2(6x+1)^3) + 0(3(6x+1)^2-2(6x+1)^3) + \frac{2{,}72211}{6}(6x+1-2(6x+1)^2+(6x+1)^3)$$

$$+ \frac{3{,}13613}{6}(-t^2+t^3) = x^3(-216+97{,}99596+112{,}90068) + x^2(-54+16{,}33266+37{,}63356)$$

$$+ 3{,}13613x = x^3(-5{,}10336) + x^2(-0{,}03378) + 3{,}13613x$$

$$\frac{dH_2(t)}{dx} = \frac{6(\frac{1}{2})}{\frac{1}{6}}((6x+1)-(6x+1)^2) + 2{,}72211(1-4(6x+1)+3(6x+1)^2) + 3{,}13613(-2(6x+1)+3(6x+1)^2)$$

$$\frac{d^2H_2(t)}{dx^2} = 108(1-2(6x+1)) + 2{,}72211(-4+6(6x+1))\cdot 6 + 6\cdot 3{,}13613(-2+6(6x+1))$$

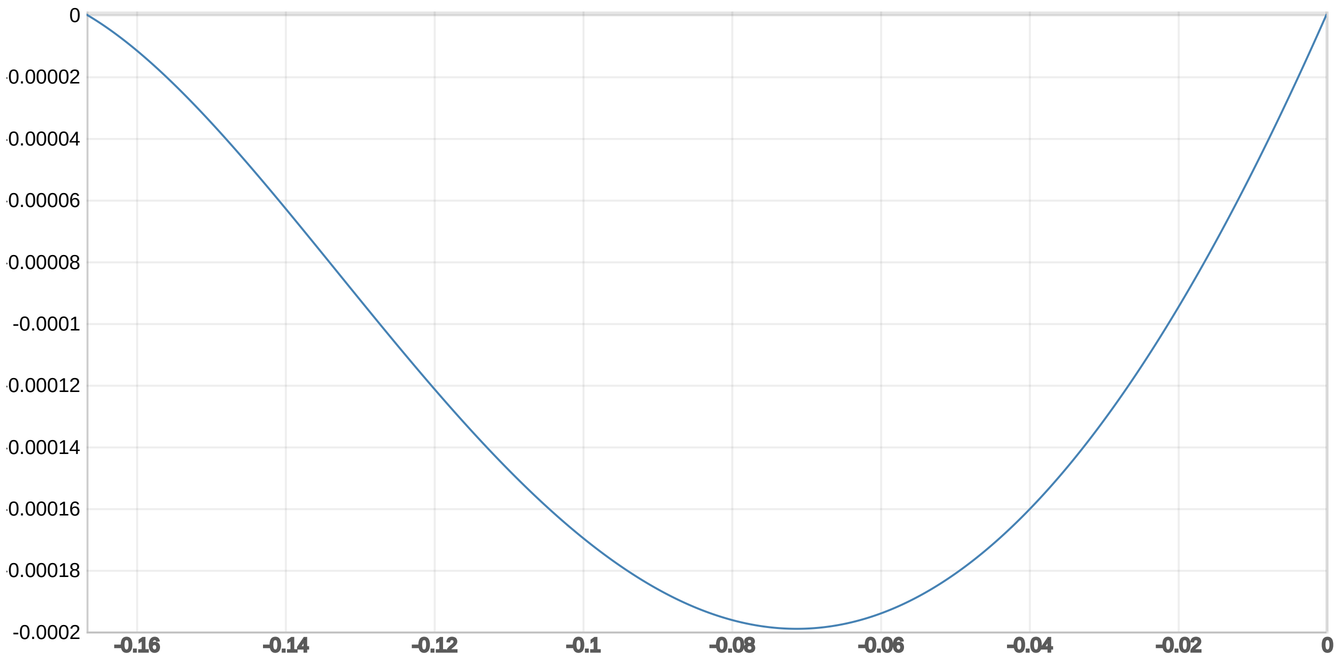
$$\frac{d^3H_2(t)}{dx^3} = -1296 + \frac{6(2{,}72211+3{,}13613)}{\frac{1}{36}}$$

$$r_2(x) = \sin(\pi x) + x^3(5{,}10336) + x^2(0{,}03378) - 3{,}13613x$$

$$\frac{dr_2(x)}{dx} = \pi\cos(\pi x) + 3x^2(5{,}10336) + 2x(0{,}03378) - 3{,}13613$$

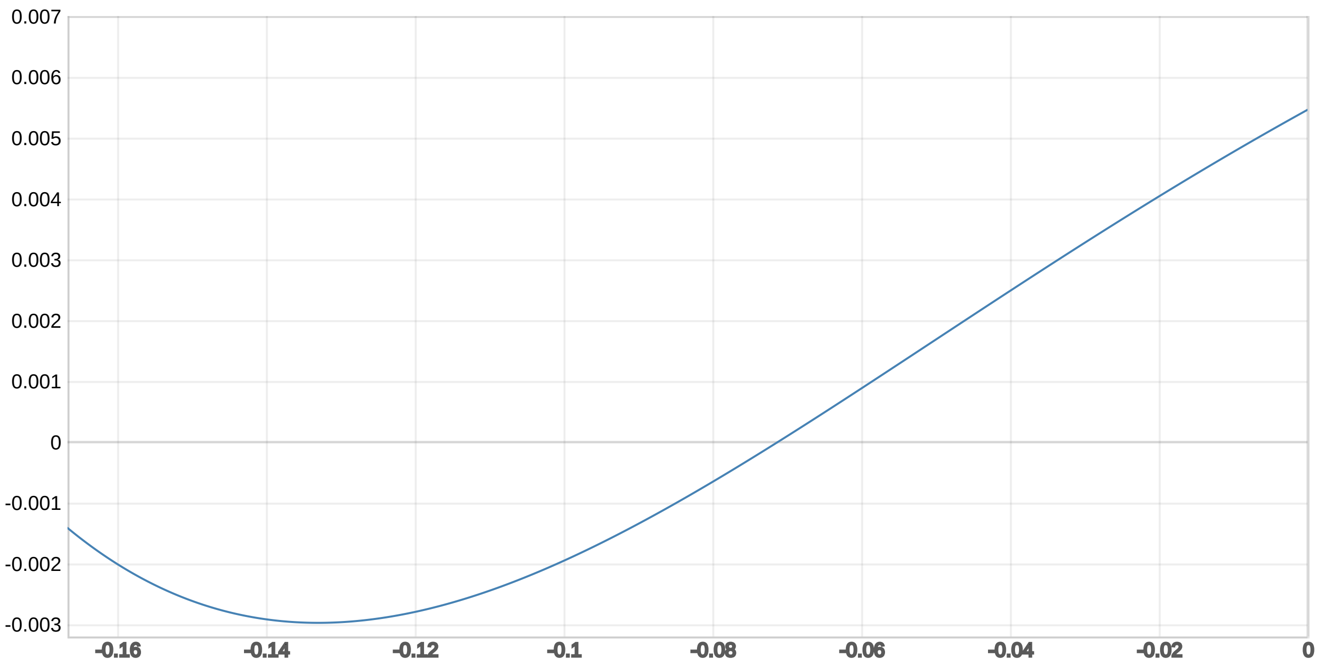
$r_2(x)$:

$$\left[5.10336 \cdot x^3 + 3.378 \times 10^{-2} \cdot x^2 - 3.13613 \cdot x + \sin\left(3.1415926536 \times 10^0 \cdot x\right)\right]$$



$$\frac{dr_2(x)}{dx}$$

$$[15.31008 \cdot x^2 + 6.756 \times 10^{-2} \cdot x + 3.1415926536 \times 10^0 \cdot \cos\left(3.1415926536 \times 10^0 \cdot x\right) - 3.1361344]$$



$$h_3 = \frac{1}{4};\ t = 4x \quad x \in \left[0; \frac{1}{4}\right]$$

$$H_3(t) = 0(1-3(4x)^2+2(4x)^3) + [illegible](3(4x)^2-2(4x)^3) + \frac{3{,}13613}{4}(4x-2(4x)^2+(4x)^3) + \frac{\pi\sqrt{2}}{8}(-(4x)^2+(4x)^3) =$$

$$= x^3(-64\sqrt{2}+50{,}17808+8\pi\sqrt{2}) + x^2(24\sqrt{2}-25{,}08904-2\pi\sqrt{2}) + 3{,}13613x$$

$$\frac{dH_3(t)}{dx} = \frac{\frac{\sqrt{2}}{2}\cdot 6}{\frac{1}{4}}(4x-(4x)^2) + 3{,}13613(1-4(4x)+3(4x)^2) + \frac{\pi\sqrt{2}}{2}(-2(4x)+3(4x)^2)$$

$$\frac{d^2H_3(t)}{dx^2} = 48\sqrt{2}(1-8x) + 3{,}13613(-4+6(4x))\cdot 4 + 2\pi\sqrt{2}(-2+6(4x))$$

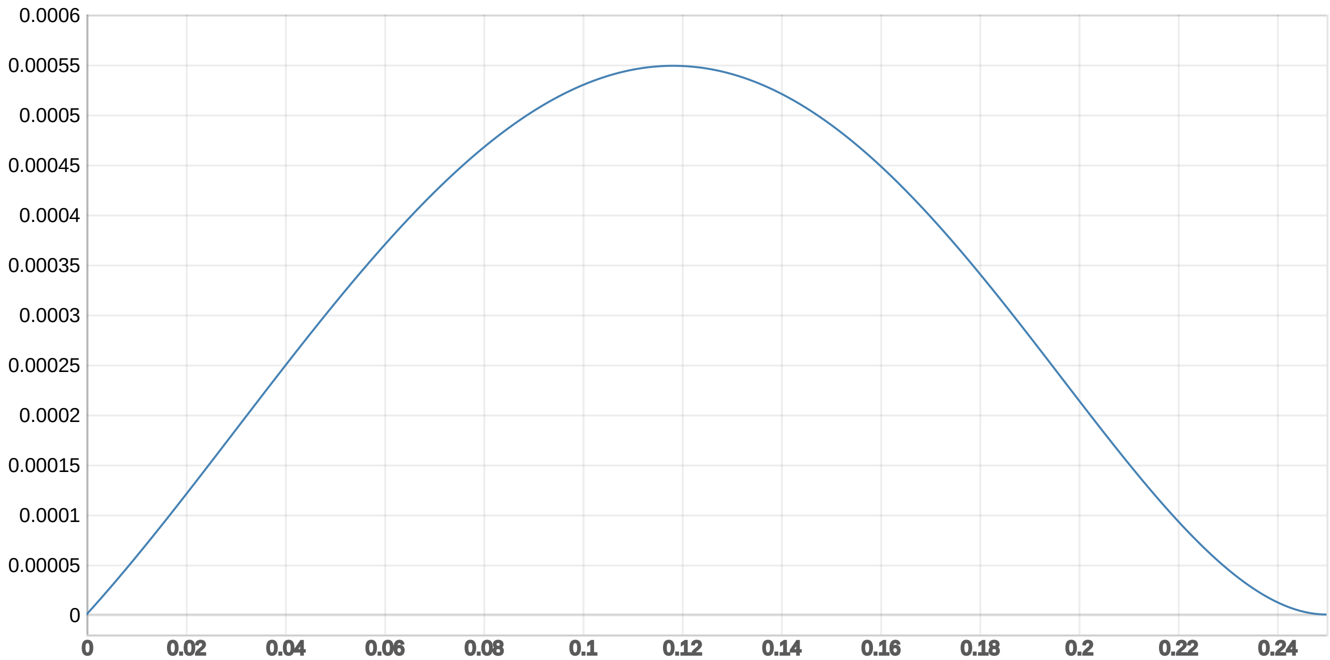
$$\frac{d^3H_3(t)}{dx^3} = -386\sqrt{2} + 96\left(3{,}13613+\frac{\pi\sqrt{2}}{2}\right)$$

$$r_3(x) = \sin(\pi x) + x^3(64\sqrt{2}-50{,}17808-8\pi\sqrt{2}) - x^2(24\sqrt{2}-25{,}08904-2\pi\sqrt{2}) - 3{,}13613x$$

$$\frac{dr_3(x)}{dx} = \pi\cos(\pi x) + 3x^2(64\sqrt{2}-50{,}17808-8\pi\sqrt{2}) - 2x(24\sqrt{2}-25{,}08904-2\pi\sqrt{2}) - 3{,}13613$$

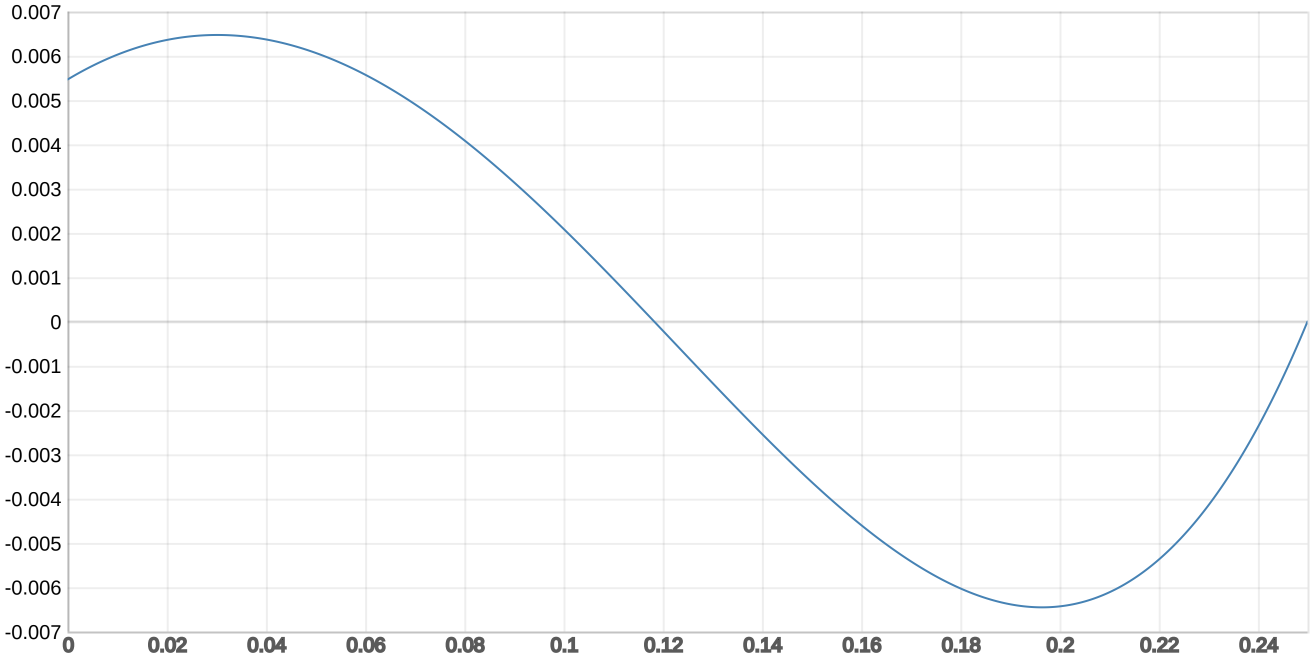
$r_3(x)$:

$$[4.788524461 \cdot x^3 + 0.033680385 \cdot x^2 - 3.13613 \cdot x + \sin\left(3.1415926536 \times 10^0 \cdot x\right)]$$

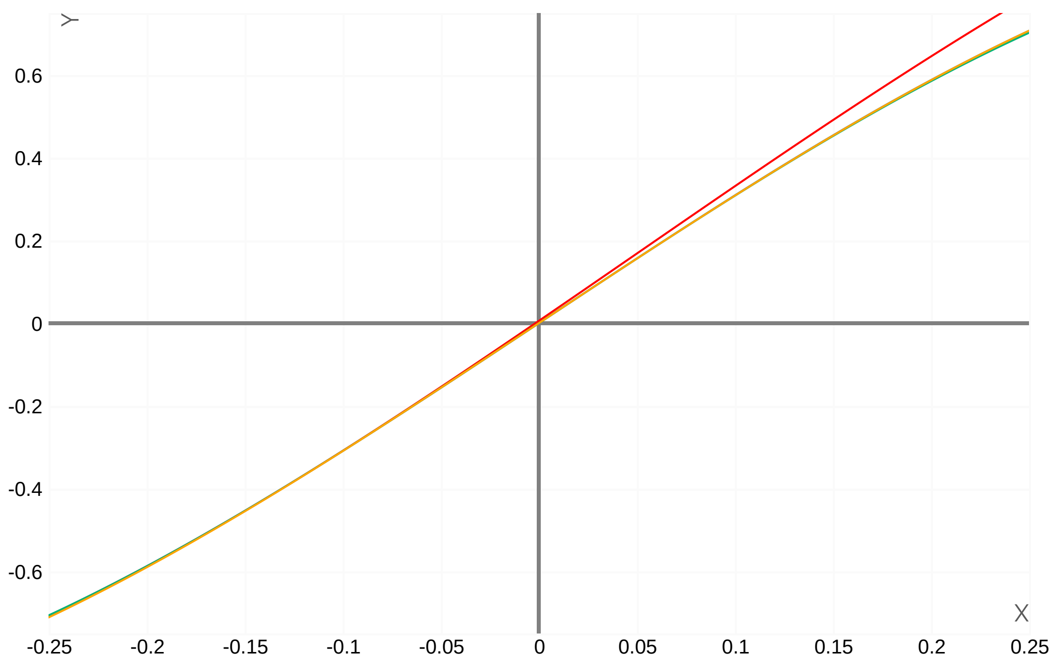


$\dfrac{dr_3(x)}{dx}$:

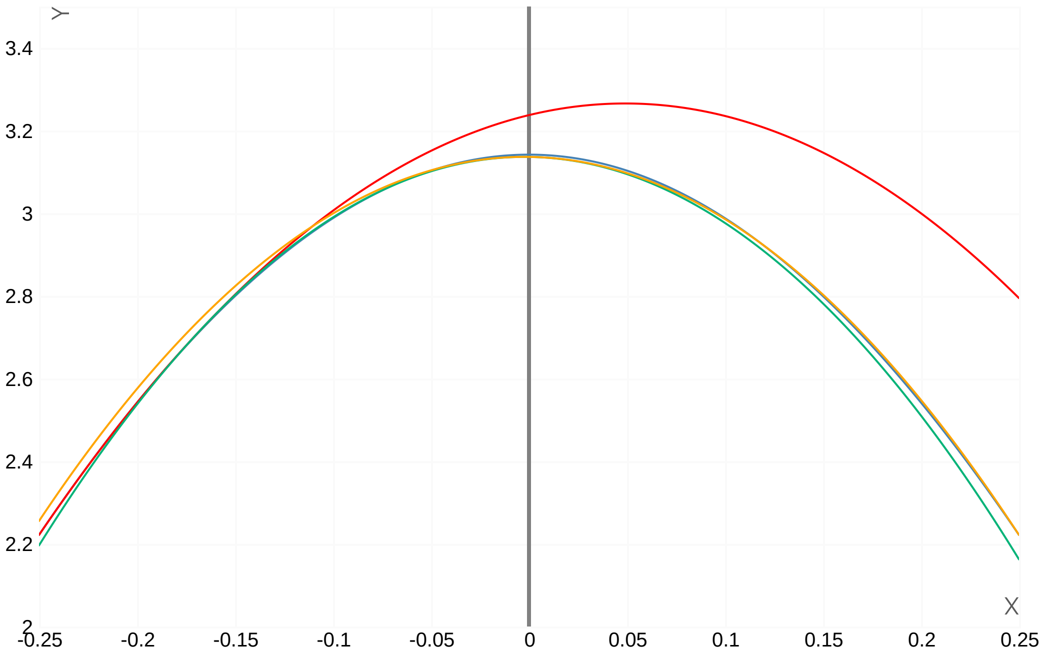
$$[14.36557338 \cdot x^2 + 0.06736077 \cdot x + 3.1415926536 \times 10^0 \cdot \cos\left(3.1415926536 \times 10^0 \cdot x\right) - 3.13613]$$



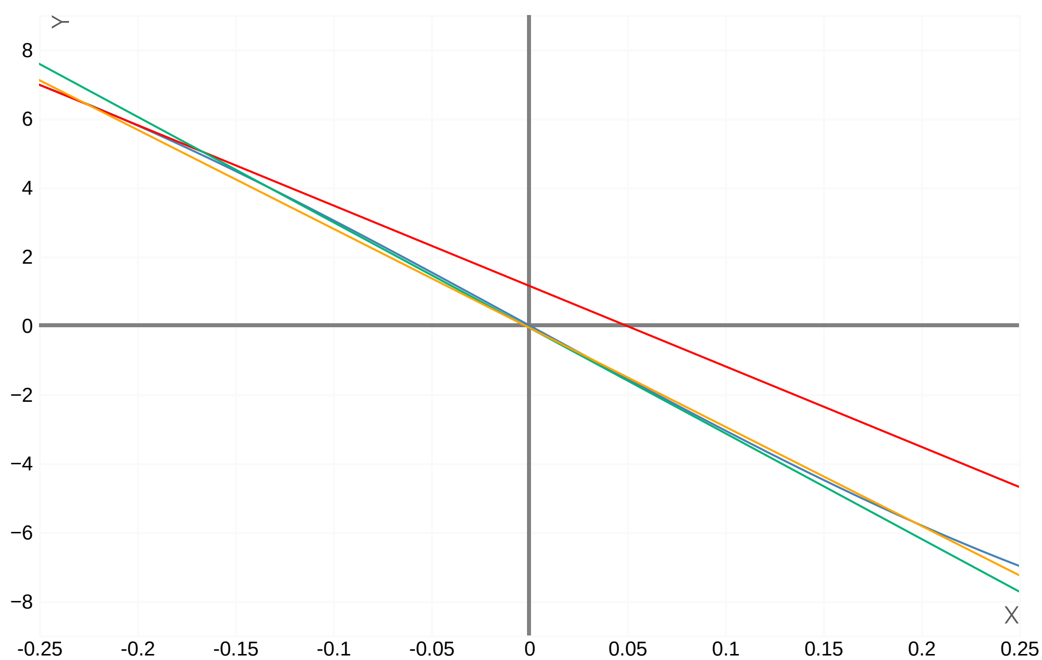
$H_1(x)$-Красный, $H_2(x)$-Зеленый, $H_3(x)$-Желтый, $sin(\pi x)$-Синий



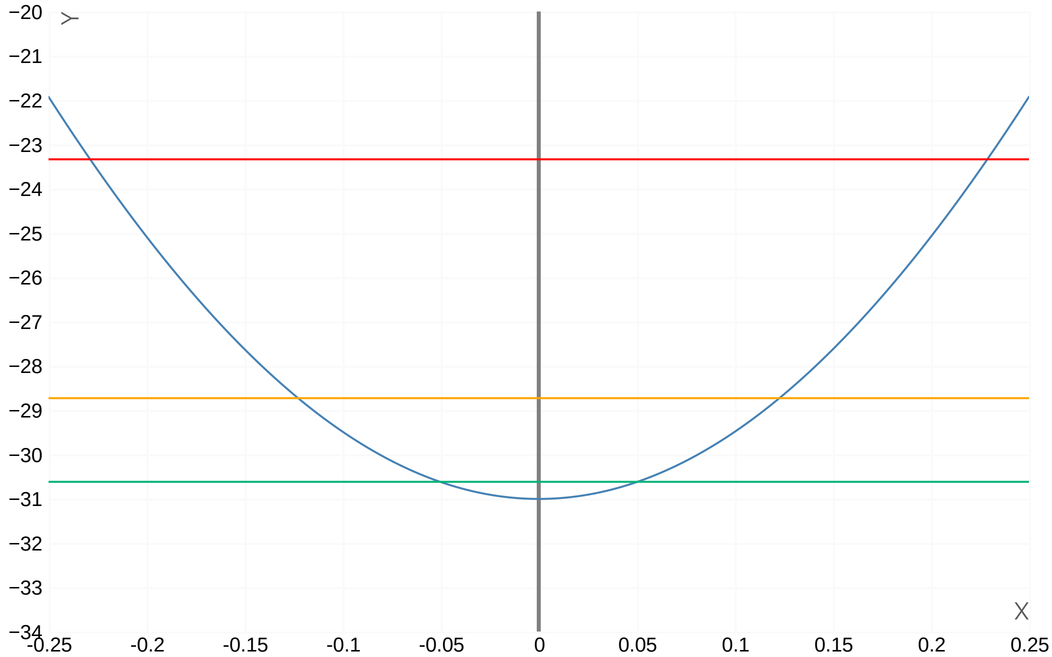
$\frac{dH_1(x)}{dx}$-Красный, $\frac{dH_2(x)}{dx}$-Зеленый, $\frac{dH_3(x)}{dx}$-Желтый, $\pi\cos(\pi x)$-Синий



$\frac{d^2H_1(x)}{dx^2}$-Красный, $\frac{d^2H_2(x)}{dx^2}$-Зеленый, $\frac{d^2H_3(x)}{dx^2}$-Желтый, $-\pi^2\sin(\pi x)$-Синий



$\frac{d^3H_1(x)}{dx^3}$-Красный, $\frac{d^3H_2(x)}{dx^3}$-Зеленый, $\frac{d^3H_3(x)}{dx^3}$-Желтый, $-\pi^3\cos(\pi x)$-Синий



Y
X
−20
−21
−22
−23
−24
−25
−26
−27
−28
−29
−30
−31
−32
−33
−34
-0.25
-0.2
-0.15
-0.1
-0.05
0
0.05
0.1
0.15
0.2
0.25